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SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4

Total Marks:40

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

MARK ALLOCATION

	Understanding of Case (25%)	Application of Concepts (25%)	Analysis (20%)	Solution Design (20%)	Clarity (10%)
Q1	2.5	2	1.5	2	1
Q2	2	2	2	2	1
Q3	2	1.5	2	1.5	0.5
Q4	2.5	2	1.5	1.5	0.5
Q5					

33.5
40

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Scenario-Based

Code of Conduct:

I __j.manoj(192572143)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

j.manoj

QUESTION 1:

A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

1. Aim

To partition the given IPv4 address block 192.168.100.0/24 into distinct variable-sized subnets for the School of Computing, School of Electronics, and School of Mechanical Engineering based on host requirements while minimizing address wastage and leaving reserved space for future expansion.

2. Given Data

- Base IP Network Block: 192.168.100.0/24 (Class C Private Range)
- Total IP Addresses Available: $2^{(32-24)} = 256$ IP addresses.
- **Departmental Allocations Required:**
- School of Computing: Allocated Subnet Mask /25
- School of Electronics: Allocated Subnet Mask /26
- School of Mechanical Engineering: Allocated Subnet Mask /27

3. Theoretical Background & Formulas

Subnetting breaks a large network into smaller manageable sub-networks. Under VLSM, subnet masks are allocated dynamically according to size requirements.

Core Subnetting Formulas

1. Total Addresses per Subnet = $2^{(32 - \text{CIDR})}$
2. Usable Host Addresses = $2^{(32 - \text{CIDR})} - 2$ (subtracting Network & Broadcast Addresses)
3. Block Size (Increment) = 256 - Subnet Mask Octet Value
4. Broadcast Address = Network Address + Block Size - 1

4. Step-by-Step Calculation

We process allocations from largest subnet mask (smallest prefix number) to smallest subnet mask to maintain contiguous binary alignment:

Step 4.1: Subnet 1 — School of Computing (/25)

- Subnet Mask: 255.255.255.128 (11111111.11111111.11111111.10000000)
- Host bits (h) = $32 - 25 = 7$. Block size = $2^7 = 128$.
- Network Address: Starts at base 192.168.100.0.
- First Usable Host: $192.168.100.0 + 1 = 192.168.100.1$.
- Broadcast Address: $192.168.100.(0 + 128 - 1) = 192.168.100.127$.
- Last Usable Host: $192.168.100.127 - 1 = 192.168.100.126$.
- Usable Hosts: $128 - 2 = 126$ hosts.

Step 4.2: Subnet 2 — School of Electronics (/26)

- Next available IP block starts immediately at 192.168.100.128.
- Subnet Mask: 255.255.255.192. Host bits (h) = $32 - 26 = 6$. Block size = $2^6 = 64$.
- Network Address: 192.168.100.128.
- First Usable Host: 192.168.100.129.
- Broadcast Address: $192.168.100.(128 + 64 - 1) = 192.168.100.191$.
- Last Usable Host: 192.168.100.190.
- Usable Hosts: $64 - 2 = 62$ hosts.

Step 4.3: Subnet 3 — School of Mechanical Engineering (/27)

- Next available IP block starts at 192.168.100.192.
- Subnet Mask: 255.255.255.224. Host bits (h) = $32 - 27 = 5$. Block size = $2^5 = 32$.
- Network Address: 192.168.100.192.
- First Usable Host: 192.168.100.193.
- Broadcast Address: $192.168.100.(192 + 32 - 1) = 192.168.100.223$.
- Last Usable Host: 192.168.100.222.
- Usable Hosts: $32 - 2 = 30$ hosts.

Step 4.4: Subnet 4 — Remaining Unused Block (/27)

- Starts at 192.168.100.224 up to 192.168.100.255 (Block size 32, 30 usable IPs reserved).

5. Subnet Allocation & Network Topology Diagram

Computing (/25) .0 to .127 (126 Hosts)	Electronics (/26) .128 to .191 (62 Hosts)	Mech (/27) .192-.223	Unused .224-.255
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6. Results Summary

Department / Subnet	Subnet Mask	Network Address	First Usable Host	Last Usable Host	Broadcast Address	Usable Hosts
School of Computing	255.255.255.128 (/25)	192.168.100.0	192.168.100.1	192.168.100.126	192.168.100.127	126

Department / Subnet	Subnet Mask	Network Address	First Usable Host	Last Usable Host	Broadcast Address	Usable Hosts
School of Electronics	255.255.255.192 (/26)	192.168.100.128	192.168.100.129	192.168.100.190	192.168.100.191	62
School of Mechanical Eng.	255.255.255.224 (/27)	192.168.100.192	192.168.100.193	192.168.100.222	192.168.100.223	30
Unused Reserved Block	255.255.255.224 (/27)	192.168.100.224	192.168.100.225	192.168.100.254	192.168.100.255	30

Conclusion: The Class C IP block 192.168.100.0/24 was successfully subnetted using VLSM principles. All three campus departments are allocated dedicated subnets with optimal host capacities without overlapping. A reserve block of 30 usable IP addresses (192.168.100.224/27) is maintained for future campus expansion, ensuring 100% address efficiency and zero address collision.

QUESTION 2:

A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

1. Aim

To allocate subnets systematically from a Class B private block 10.10.0.0/16 to fulfill structural demands across four core enterprise departments: Software Development (/24), Quality Assurance (/25), Human Resources (/26), and Executive Management (/27).

2. Given Data

- Base Private IP Block: 10.10.0.0/16
- Total IP Capacity: $2^{16} = 65,536$ IP addresses.

- **Requested Prefix Allocations:**
- Software Development: /24
- Quality Assurance (QA): /25
- Human Resources (HR): /26
- Executive Management: /27

3. Theoretical Background & Formulas

When partitioning a large address space like /16, prefix lengths determine how subnets split across third and fourth octets.

Mathematical Subnet Formulas

Subnet Capacity = $2^{(32 - \text{Prefix})}$ Fourth Octet Boundary Increment = $256 - \text{SubnetMaskOctet4}$
 Next Subnet Start = Current Network IP + Block Size

4. Step-by-Step Calculation

Step 4.1: Software Development Subnet (/24)

- Subnet Mask: 255.255.255.0. Host bits = $32 - 24 = 8$. Block size = $2^8 = 256$.
- Network Address: 10.10.0.0.
- First Host: 10.10.0.1. Last Host: 10.10.0.254.
- Broadcast Address: 10.10.0.255. Usable Hosts: 254.

Step 4.2: Quality Assurance (QA) Subnet (/25)

- Next available address: 10.10.1.0.
- Subnet Mask: 255.255.255.128. Host bits = $32 - 25 = 7$. Block size = $2^7 = 128$.
- Network Address: 10.10.1.0.
- First Host: 10.10.1.1. Last Host: 10.10.1.126.
- Broadcast Address: 10.10.1.127. Usable Hosts: 126.

Step 4.3: Human Resources (HR) Subnet (/26)

- Next available address: 10.10.1.128.
- Subnet Mask: 255.255.255.192. Host bits = $32 - 26 = 6$. Block size = $2^6 = 64$.
- Network Address: 10.10.1.128.
- First Host: 10.10.1.129. Last Host: 10.10.1.190.
- Broadcast Address: 10.10.1.191. Usable Hosts: 62.

Step 4.4: Executive Management Subnet (/27)

- Next available address: 10.10.1.192.
- Subnet Mask: 255.255.255.224. Host bits = $32 - 27 = 5$. Block size = $2^5 = 32$.
- Network Address: 10.10.1.192.
- First Host: 10.10.1.193. Last Host: 10.10.1.222.
- Broadcast Address: 10.10.1.223. Usable Hosts: 30.

5. Enterprise Hierarchy Subnet Map Diagram

Enterprise Block: 10.10.0.0/16 (65,536 IPs)

Software Dev (/24) 10.10.0.0 - 10.10.0.255	QA (/25) 10.10.1.0 - .127	HR (/26) .128 - .191	Exec (/27) .192 - .223
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6. Results Summary

Department	Subnet Mask	Network Address	First Usable Host	Last Usable Host	Broadcast Address	Usable Hosts
Software Dev	255.255.255.0 (/24)	10.10.0.0	10.10.0.1	10.10.0.254	10.10.0.255	254
QA Department	255.255.255.128 (/25)	10.10.1.0	10.10.1.1	10.10.1.126	10.10.1.127	126
Human Resources	255.255.255.192 (/26)	10.10.1.128	10.10.1.129	10.10.1.190	10.10.1.191	62
Exec Management	255.255.255.224 (/27)	10.10.1.192	10.10.1.193	10.10.1.222	10.10.1.223	30
Unused Subnet Space	Dynamic Block	10.10.1.224	10.10.1.225	10.10.255.254	10.10.255.255	65,054

Conclusion: The Enterprise address allocation systematically assigns address blocks tailored to each department's scale. By starting allocations at 10.10.0.0, the layout retains thousands of addresses (10.10.1.224 onwards) in reserve, eliminating subnet fragmentation and allowing clean route summarization.

QUESTION 3:

A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

1. Aim

To analyze, compress, expand, and determine network prefix metrics for the smart hospital's assigned IPv6 block 2001:0db8:abcd:0000:0000:0000:0000/64 in compliance with RFC 5952 recommendations.

2. Given Data

- Assigned Uncompressed IPv6 Address: 2001:0db8:abcd:0000:0000:0000:0000/64
- Prefix Length: /64 (Standard Global Unicast Interface Prefix)
- Total Address Space Size: 128-bit address architecture.

3. Theoretical Background & Standards (RFC 5952)

IPv6 addresses consist of 128 bits expressed as eight 16-bit hexadecimal blocks separated by colons. RFC 5952 defines canonical formatting rules:

IPv6 Standard Rules

1. Leading Zero Omission: Omit leading zeros in any individual 16-bit field (e.g., 0db8 -> db8). 2. Double Colon Compression (::): Replace longest single contiguous sequence of all-zero fields with :: (allowed only once). 3. Host Address Range Capacity: $2^{(128 - \text{Prefix})} = 2^{64}$ total unique Interface Identifiers per subnet.

4. Step-by-Step Calculation & Compression

Step 4.1: Rule 1 — Omit Leading Zeros

- Original 8 groups: 2001 : 0db8 : abcd : 0000 : 0000 : 0000 : 0000 : 0000
- Removing leading zeros in each 16-bit block yields: 2001 : db8 : abcd : 0 : 0 : 0 : 0 : 0

Step 4.2: Rule 2 — Apply Double Colon (::) Compression

- There are 5 consecutive all-zero blocks from 4th to 8th group (0:0:0:0:0). Replacing this sequence with :: produces the canonical compressed format: 2001:db8:abcd::/64.

Step 4.3: Calculate Total Interface Addresses

- Interface Identifier bits = 128 - 64 = 64 bits.
- Total addresses = $2^{64} = 18,446,744,073,709,551,616$ addresses.

5. IPv6 128-Bit Structure Diagram



5. Results Summary

IPv6 Parameter	Formatted Representation	Engineering Explanation
Canonical Compressed Notation	2001:db8:abcd::/64	Compliant with RFC 5952 (leading zeros stripped & 5 zero-blocks compressed to ::).
Expanded Hexadecimal Representation	2001:0db8:abcd:0000:0000:0000:0000:0000	Fully expanded 128-bit structure showing all 8 16-bit fields.
Global Routing & Subnet Prefix	2001:0db8:abcd:0000::/64	First 64 bits representing network prefix identifier.

IPv6 Parameter	Formatted Representation	Engineering Explanation
Total Host Capacity	18,446,744,073,709,551,616	2 ⁶⁴ unique host endpoints supportable without sub-delegation.

Conclusion: The smart hospital's IPv6 address block was properly formatted into its canonical compressed representation 2001:db8:abcd::/64. The /64 prefix provides 2⁶⁴ Interface Identifiers, guaranteeing virtually infinite scaling for medical IoT devices, telemetry monitoring systems, and health record servers without requiring NAT.

QUESTION 4:

An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

1. Aim

To analyze an incoming IP packet destination address (192.168.2.45) against an ISP core router forwarding table and determine the exact matching route entry, next-hop forwarding interface, and default routing mechanisms.

2. Given Data

- **Incoming Destination IP Packet: 192.168.2.45**
- **Active Router Forwarding Table Entries:**
 - Entry A: 192.168.1.0/24 (Interface eth0)
 - Entry B: 192.168.2.0/24 (Interface eth1)
 - Entry C: 192.168.3.0/24 (Interface eth2)
 - Entry D: 0.0.0.0/0 (Default Gateway - Interface eth3)

3. Theoretical Background & Longest Prefix Match (LPM)

Layer 3 routers evaluate routing tables using bitwise AND operations between the destination IP address and subnet masks of each entry, selecting the matching prefix with the longest bit-length (most specific match).

Routing Lookup Logic

Matching Network = Destination IP AND Subnet Mask. If multiple matches exist, the entry with the highest prefix value (e.g., /24 over /16) is selected.

4. Step-by-Step Bitwise Calculation

Step 4.1: Convert Destination IP to Binary

- Destination 192.168.2.45 in Binary: 11000000 . 10101000 . 00000010 . 00101101

Step 4.2: Evaluate Entry B — 192.168.2.0/24

- Subnet Mask 255.255.255.0 Binary: 11000000 . 10101000 . 00000010 . 00000000
- Bitwise AND Result: 192.168.2.0
- Match Status: EXACT MATCH (24 bits matched).

Step 4.3: Evaluate Other Entries

- Entry A (192.168.1.0/24): AND result = 192.168.2.0 ≠ 192.168.1.0 (No Match).
- Entry C (192.168.3.0/24): AND result = 192.168.2.0 ≠ 192.168.3.0 (No Match).
- Entry D (0.0.0.0/0): Matches 0 bits (Default fallback only).

5. Core Router Forwarding Diagram

Incoming Packet Dst: 192.168.2.45	192.168.1.0/24 (eth0) no match	192.168.2.0/24 (eth1) MATCH -> forwarded here	192.168.3.0/24 (eth2) no match
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6. Results Summary

Routing Evaluation Metric	Determined Result	Technical Reasoning
Destination Address Analyzed	192.168.2.45	Target IPv4 packet header address extract.
Target Subnet Range	192.168.2.0 – 192.168.2.255	Calculated host range encompassing the packet's destination.
Selected Routing Entry	192.168.2.0/24 (Entry B)	Highest matching prefix length (24 matching binary network bits).
Outgoing Next-Hop Interface	Interface eth1	Directly egress port connected to the destination LAN segment.
Default Route Role	0.0.0.0/0 (Unused)	Default gateway is bypassed because an explicit match (/24) exists.

Conclusion: The incoming packet destined for 192.168.2.45 was correctly evaluated using Longest Prefix Match logic. It matched the 192.168.2.0/24 routing table entry across 24 contiguous bits, causing the router to forward the packet out of interface eth1.

QUESTION 5:

A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

1. Aim

To design an IP addressing scheme and Layer-3 router interface layout enabling bi-directional inter-subnet routing between the Administration Department LAN (192.168.10.0/26) and Finance Department LAN (192.168.10.64/26).

2. Given Data

- Admin LAN Subnet: 192.168.10.0/26 | Default Gateway IP: 192.168.10.1
- Finance LAN Subnet: 192.168.10.64/26 | Default Gateway IP: 192.168.10.65
- Subnet Mask for Both LANs: 255.255.255.192 (/26)
- Requirement: Assign static host IPs for 3 computers in each LAN.

3. Theoretical Background & Formulas

Layer-3 inter-subnet communication requires hosts in different IP subnets to route packets through their local default gateway (the L3 router interface IP).

Subnet Boundary Calculations

Subnet Mask /26 = 255.255.255.192 Block Size = 256 - 192 = 64 Usable IP Range = (Network Address + 1) to (Broadcast Address - 1)

4. Step-by-Step Subnet & IP Mapping

Step 5.1: Administration LAN Breakdown

- Network Address: 192.168.10.0 | Broadcast Address: 192.168.10.63
- Router Gateway Interface (eth0): 192.168.10.1

- Valid Host Range: 192.168.10.1 to 192.168.10.62
- Computer 1 IP: 192.168.10.2 | Computer 2 IP: 192.168.10.3 | Computer 3 IP: 192.168.10.4

Step 4.2: Finance LAN Breakdown

- Network Address: 192.168.10.64 | Broadcast Address: 192.168.10.127
- Router Gateway Interface (eth1): 192.168.10.65
- Valid Host Range: 192.168.10.65 to 192.168.10.126
- Computer 1 IP: 192.168.10.66 | Computer 2 IP: 192.168.10.67 | Computer 3 IP: 192.168.10.68

5. Inter-Department L3 Topology Diagram

ADMIN LAN (/26) Net .0 PCs .2 .3 .4 GW eth0 .1	L3 ROUTER eth0 (.1) <-> eth1 (.65)	FINANCE LAN (/26) Net .64 PCs .66 .67 .68 GW eth1 .65
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6. Results Summary

Department LAN	Network Addr	Broadcast Addr	Valid Host IP Range	Suggested Host IPs (3 PCs)	Default Gateway IP
Administration LAN	192.168.10.0	192.168.10.63	192.168.10.1 – 192.168.10.62	192.168.10.2, .3, .4	192.168.10.1 (eth0)
Finance LAN	192.168.10.64	192.168.10.127	192.168.10.65 – 192.168.10.126	192.168.10.66, .67, .68	192.168.10.65 (eth1)

Conclusion: The Layer-3 router configuration establishes isolated broadcast domains for Administration and Finance while maintaining seamless inter-departmental routing. Assigning default gateways (192.168.10.1 and 192.168.10.65) ensures host traffic is forwarded correctly across subnets.